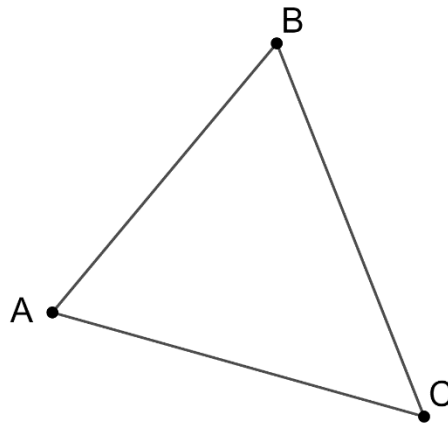


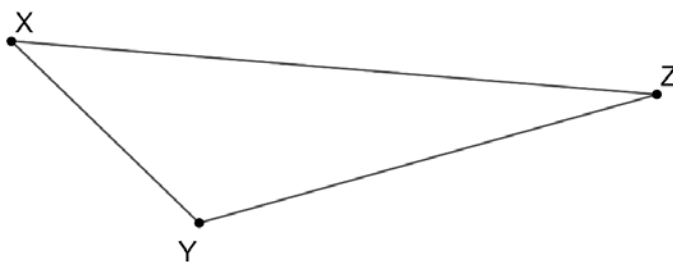
Use $\triangle ABC$ to answer questions 1-7.

1. Construct the three perpendicular bisectors for $\triangle ABC$.



2. Label the circumcenter of the triangle as Point R .
3. Use a ruler to measure the length of segment AR .
4. Use a ruler to measure the length of segment CR .
5. Use a ruler to measure the length of segment BR .
6. Use a compass to draw the circumscribed circle around the triangle.

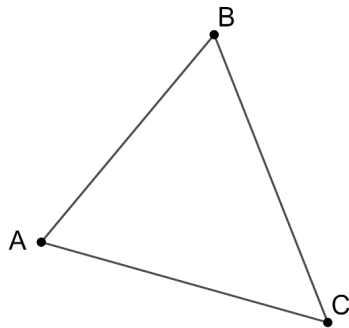
Use $\triangle XYZ$ to answer questions 7-10.



7. Construct the three perpendicular bisectors for $\triangle XYZ$.
8. Label the circumcenter of the triangle as Point T .
9. Use a compass to draw the circumscribed circle.
10. Given that the measure of arc $XYZ = 140^\circ$, what is the measure of $\angle Y$?

Use $\triangle ABC$ to answer questions 1-5.

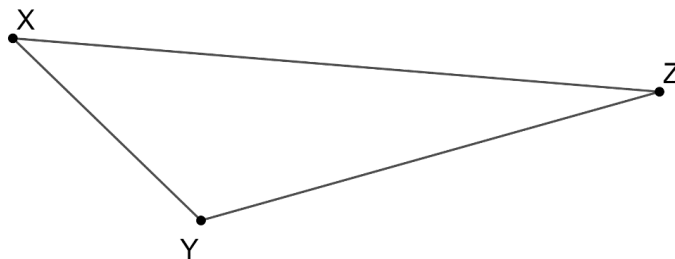
1. Construct the three angle bisectors for $\triangle ABC$.



2. Label the incenter of the triangle as Point Z .
3. Use a compass to draw the inscribed circle inside the triangle, with point Z as its center.
4. Label the three points where the inscribed circle intersects the triangle as F , G , and H .
5. Measure the lengths of segments $\overline{FZ}$, $\overline{GZ}$, and $\overline{HZ}$.

Use $\triangle XYZ$ to answer questions 6-10.

6. Construct the three angle bisectors for $\triangle XYZ$.



7. Label the incenter of the triangle as Point K .
8. Use a compass to draw the inscribed circle inside the triangle, with point K as its center.
9. Label the three points where the inscribed circle intersects the triangle as M , P , and T .
10. Measure the lengths of segments $\overline{MK}$, $\overline{PK}$, and $\overline{TK}$.

Geometry

Unit 11

Lesson 3 Homework

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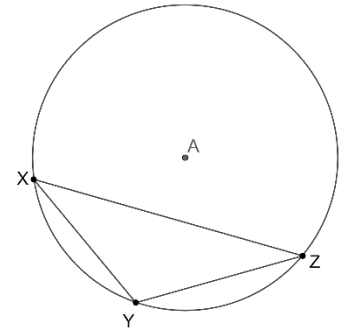
Circle A is shown, where points X, Y , and Z are on circle A and create $\triangle XYZ$. $m\widehat{XZ} = 240^\circ$, $m\angle ZXY = 20^\circ$, and $m\angle XYZ = (4x - 28)^\circ$.

1. Find the value of x .

2. Find the $m\angle XYZ$.

3. Find the $m\widehat{YZ}$.

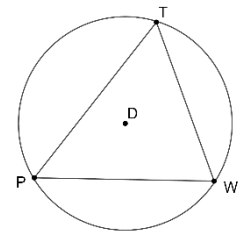
4. Find $m\angle YZX$.



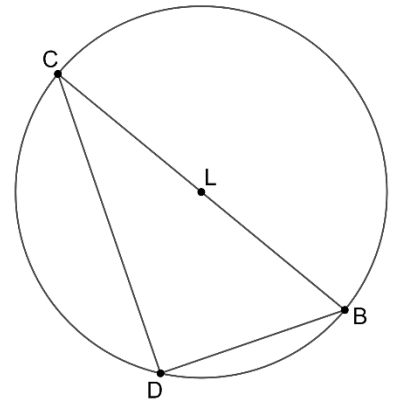
Circle D is shown, where points P, T , and W are on circle D and create $\triangle PTW$. $m\angle PTW = 54^\circ$ and $m\angle TWP = 60^\circ$.

5. Find $m\widehat{PT}$.

6. Find $m\widehat{TW}$.



Circle L has points B, C , and D on circle L and diameter $\overline{BC}$. $\triangle BCD$ is inscribed in the circle. $m\widehat{CD} = 100^\circ$, $m\angle DCB = 40^\circ$, and $m\angle DBC = (2x + 12)^\circ$.



7. What is the $m\angle CDB$?

8. What is the value of x ?

9. Determine $m\angle DBC$.

10. Determine $m\widehat{DBC}$?

Geometry

Unit 11

Lesson 4 Homework

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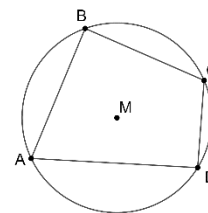
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Circle M is shown, with $ABCD$ inscribed in the circle.

$m\widehat{ADC} = 184^\circ$ and $m\angle DAB = 44^\circ$.

1. Find the $m\angle ABC$.



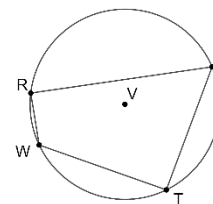
2. Find the $m\angle BCD$.

3. Find the $m\angle CDA$.

Circle V is shown, with $RSTW$ inscribed in the circle.

$m\widehat{WTS} = 190^\circ$ and $m\angle WTS = (5x + 10)^\circ$.

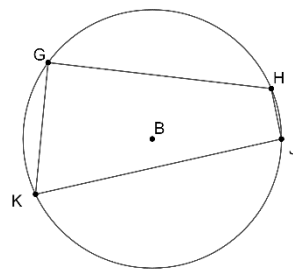
4. Find the value of x .



5. Find the $m\angle WTS$.

Circle B is shown, with $GHJK$ inscribed in the circle. $m\angle HGK = (5x + 25)^\circ$, $m\angle GKJ = (3x)^\circ$, and $m\angle KJH = (4x + 20)^\circ$.

6. Find the value of x .



7. Find $m\angle HGK$.

8. Find $m\angle GKJ$.

9. Find $m\angle KJH$.

10. Find $m\angle JHG$.

Geometry

Unit 11

Lesson 5 Homework

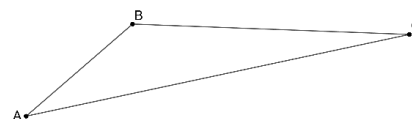
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For questions 1 through 4, $\triangle ABC$ is shown, where $m\angle A = (3x)^\circ$, $m\angle B = (7x + 10)^\circ$, and $m\angle C = (2x + 10)^\circ$.

- Find the value of x .



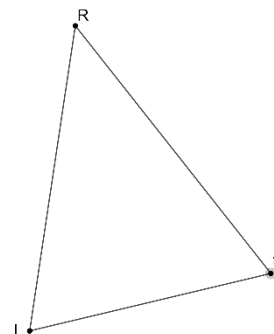
- What is the $m\angle A$?

- What is the $m\angle B$?

- What is the $m\angle C$?

For questions 5 and 6, $\triangle LRT$ is shown, where $m\angle L = m\angle T = (5x)^\circ$ and $m\angle R = (2x)^\circ$.

- Find the value of x .

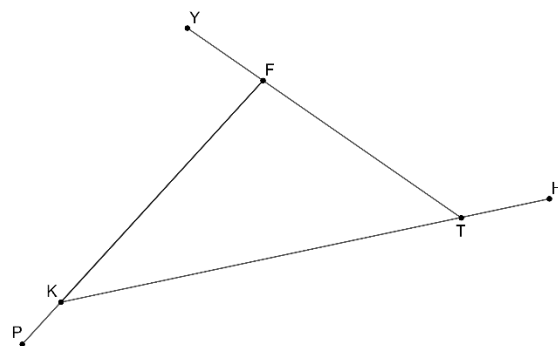


- What is the $m\angle L$?

For questions 7 through 10, $\triangle FKT$ is shown, where $\angle KFY$, $\angle PKT$, and $\angle FTH$ are exterior angles.

$m\angle FTH = (8x + 10)^\circ$, $m\angle TFK = 88^\circ$, and

$m\angle PKT = (16x - 30)^\circ$.



7. Find the value of x .

8. Find $m\angle KTF$.

9. Find $m\angle PKT$.

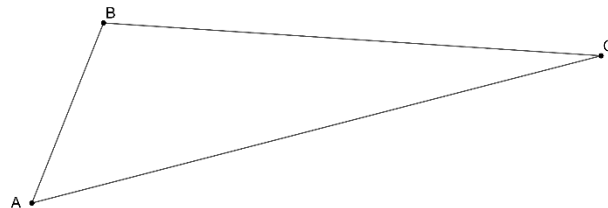
10. Find $m\angle YFK$.

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Lesson 6 Homework

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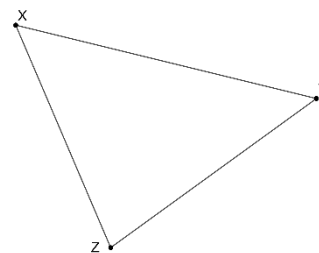
1. $\triangle ABC$ is shown, where $AB = 4$, $BC = 8$, and $AC = 10$.

List the angles in order from least to greatest.



$\triangle XYZ$ is shown. Use the triangle to complete questions 2 through 4.

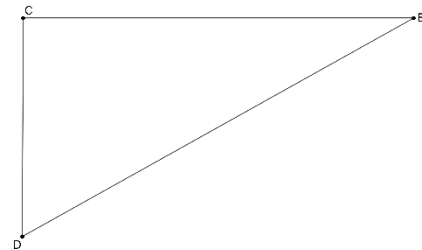
2. Use a protractor to measure each of the angles.



3. List the angles in order from smallest to largest.
4. List the side lengths in order from smallest to largest.

5. $\triangle BCD$ is shown, where $m\angle B = 30^\circ$, $m\angle D = 60^\circ$, and $m\angle C = 90^\circ$.

List the side lengths in order from largest to smallest.



For questions 6 through 8, determine whether the given side lengths can form a triangle.

6. 12 cm, 14 cm, and 20 cm

7. 2 in, 3 in, and 5 in

8. 3 ft, 8 ft, and 12 ft

For questions 9 and 10, two sides of a triangle are given. Determine a possible side length for the third side.

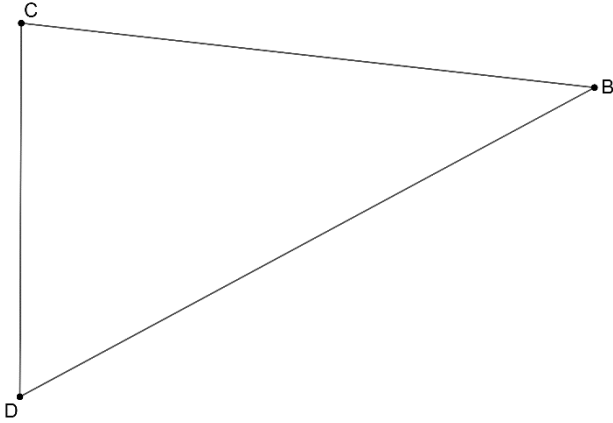
9. 7 in and 5 in

10. 10 m and 2 m

Geometry
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Lesson 7 Homework

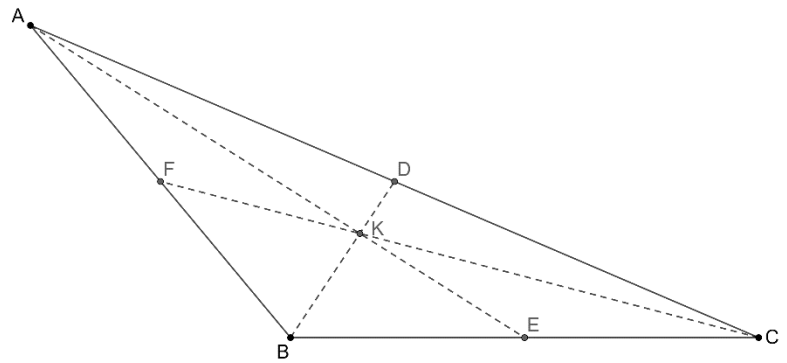
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$\triangle BCD$ is shown. Complete the steps for finding the centroid of the circle. Then, label the centroid Point Y .

Steps	Construction
Construct the perpendicular bisector of BC to find its midpoint.	5. 
1.	
2.	
Draw the median, R , to the opposite vertex, $\angle DBC$.	
3.	
4.	
Draw Point Y , the point of concurrency, where the three medians intersect.	

Use this information to answer questions 6 through 8.

Triangle ABC is shown, where point K is the centroid of the triangle. $FC = 6$, $BK = 2.4$, and $KE = 3.7$.



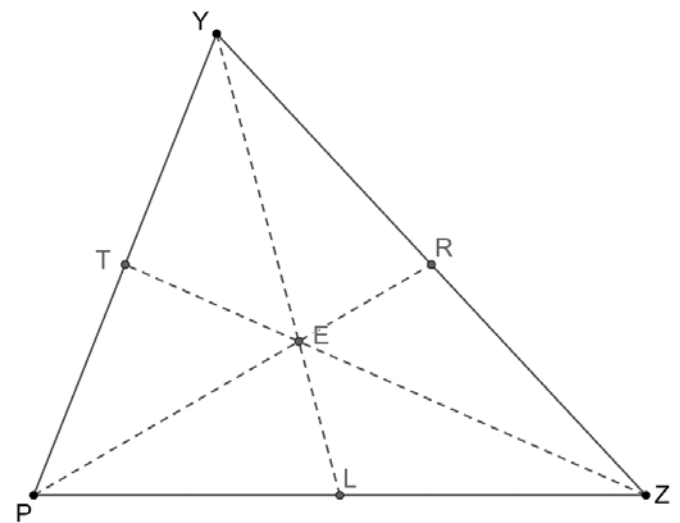
6. Find the length of $\overline{DK}$.

7. Find the length of $\overline{AE}$.

8. Find the length of $\overline{KC}$.

Use this information to answer questions 9 and 10.

Triangle YPZ is shown, where point E is the centroid of the triangle. $TZ = 6x + 9$, and $TE = 4$.



9. Find the value of x .

10. Find the length of $\overline{EZ}$.

Geometry

Unit 11

Lesson 8 Homework

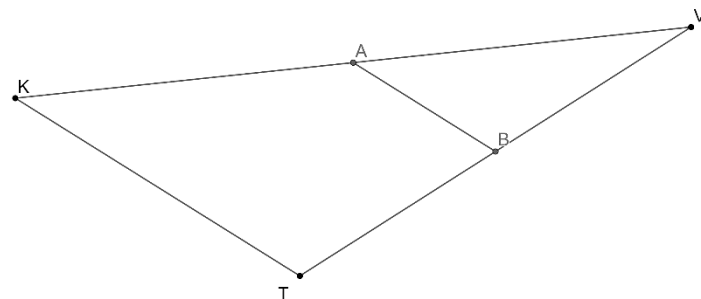
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Use this information to answer questions 1 through 5.

Triangle KVT is shown, where $\overline{AB}$ is the midsegment. $KT = 2x + 4$, $AB = \frac{1}{2}x + 3$, $AV = \frac{5}{2}y - 6.5$, $KA = \frac{1}{2}y + 3.5$, $VT = 3z - 7$, and $TB = 2x + 2.1$.



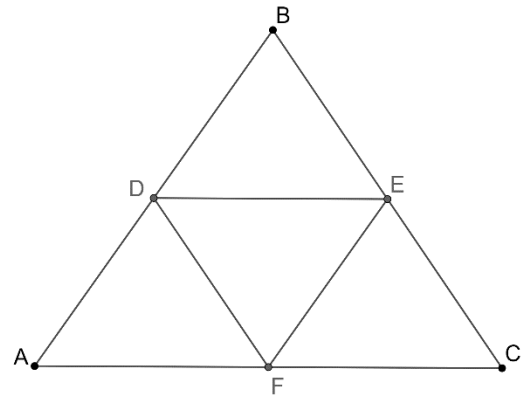
1. Find the value of x .
2. Find the value of y .
3. Find the value of z .
4. Find the length of the midsegment, $\overline{AB}$.
5. Find the length of $\overline{KV}$.

Use this information to answer questions 6 through 10.

Triangle ABC is shown, where D , E , and F are midpoints.

$DE = 2x - 4$, $AC = 4x - 8$, $DF = x + 3$, $BC = 3x + 1$,

$FE = 3x - 6$, and $AB = 5x - 7$.



6. Find the value of x .

7. Find the length of $\overline{DF}$.

8. Find the length of $\overline{EF}$.

9. Find the length of $\overline{DE}$.

10. Find the perimeter of triangle ABC .

Geometry

Unit 11

Lesson 9 Homework

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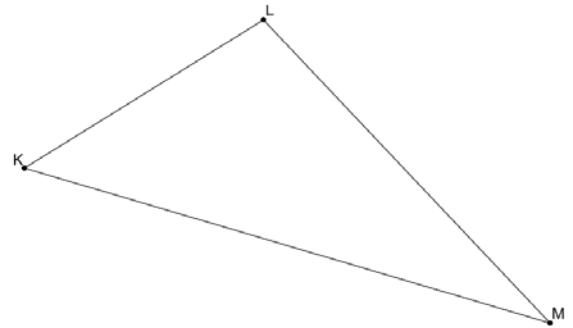
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Use this information to answer questions 1 through 5.

Triangle KLM is shown, where

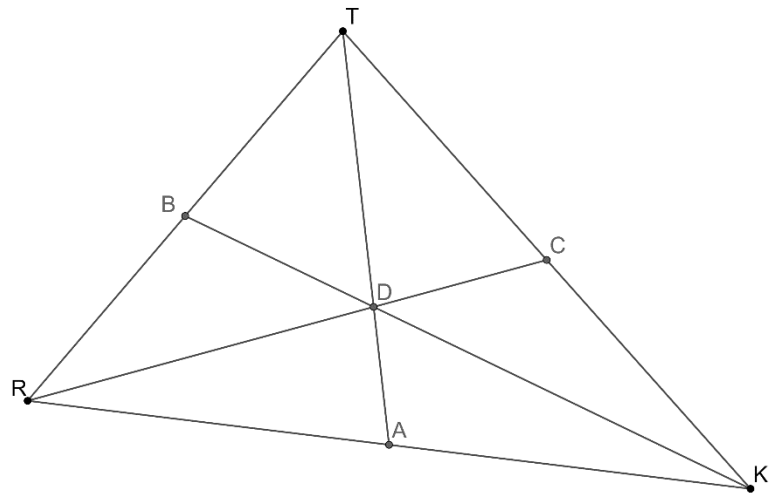
$m\angle K = (2x + 3)^\circ$, $m\angle L = (3x + 7)^\circ$, and $m\angle M = (x + 2)^\circ$.



1. Find the value of x .
2. Identify the measure of $m\angle K$.
3. Identify the measure of $m\angle L$.
4. Identify the measure of $m\angle M$.
5. List the sides of the triangle in order of their lengths, smallest to largest.

Use this information to answer questions 6 through 10.

Triangle RTK is shown, where $\overline{RC}$, $\overline{BK}$, and $\overline{AT}$ are median. $BT = 3$, $BK = 12$, $DC = 4$, $TD = 6$, and $RK = 16$.



6. Find the length of $\overline{BD}$.

7. Find the length of $\overline{RC}$.

8. Find the length of $\overline{DA}$.

9. Find the length of $\overline{RT}$.

10. Find the perimeter of triangle RBD .

Geometry

Unit 11

Lesson H1 Homework

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Complete the missing steps showing one way to construct an equilateral triangle inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	
3	Mark any point on the circumference of the circle.
4	
5	Continue around the circle making a total of five small arcs.
6	
7	

Use Point A to construct an equilateral triangle inscribed inside a circle.

A

Complete the missing steps showing one way to construct a hexagon inscribed inside a circle.

Step	Procedure
1	Use a compass to draw a circle. Mark the center and label it.
2	
3	Mark any point on the circumference of the circle.
4	
5	Continue around the circle making a total of five small arcs.
6	
7	

Use Point *P* to construct a hexagon inscribed inside a circle.

